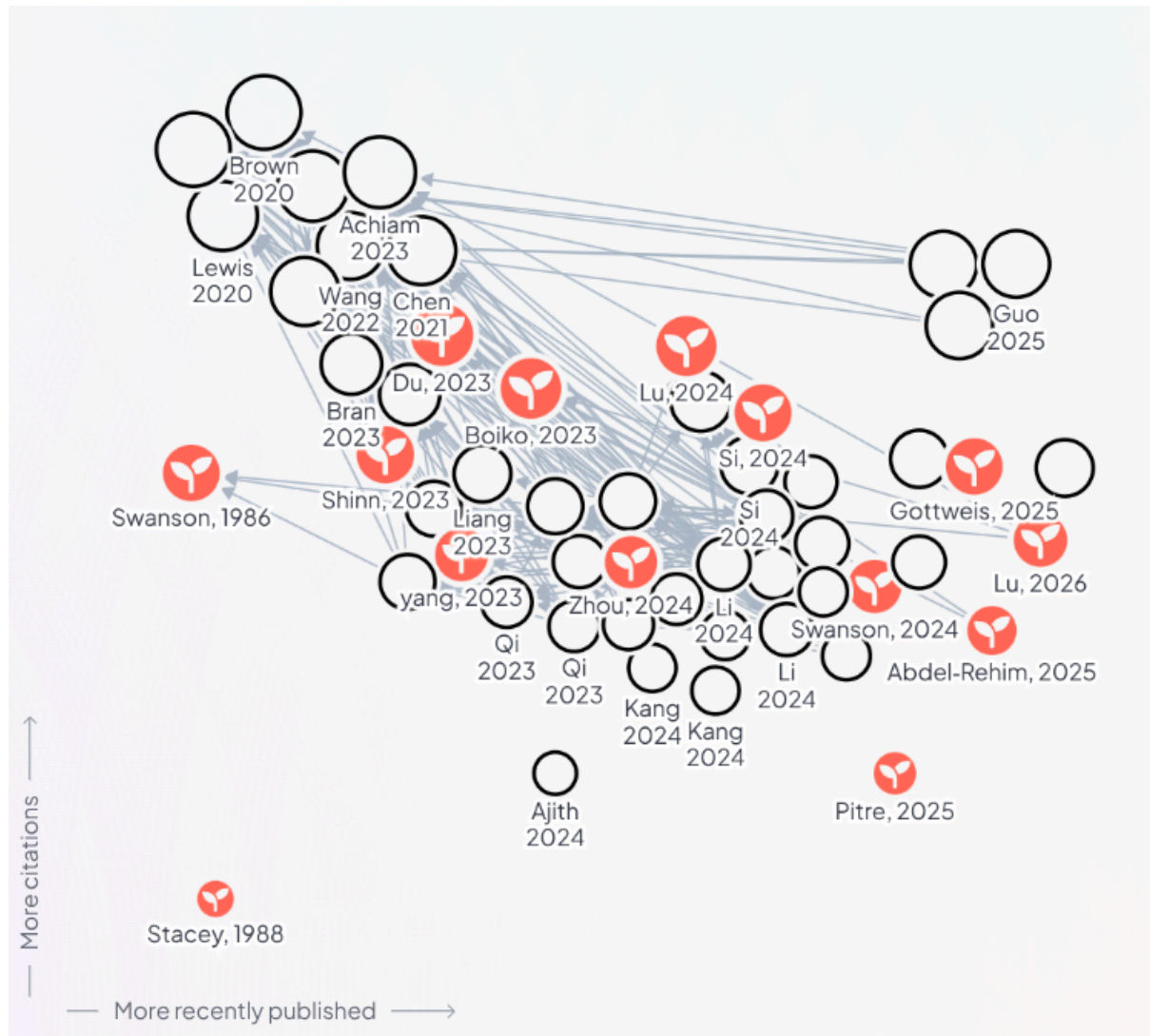


Topic: Single-LLM Versus Multi-Agent Debate Systems for Scientific Hypothesis Generation

Citation Network (Research Rabbit)



Comments:

The citation map showed relationships that were not obvious from the spreadsheet. While the spreadsheet lists the papers individually, the map shows that several papers form a dense cluster around LLM-based scientific discovery and multi-agent systems. Papers such as Zhou (2024), Si (2024), Lu (2024), Gottweis (2025), and related recent work are closely connected, suggesting that they are part of the main research area. In contrast, older papers such as Swanson (1986) and Stacey (1988) appear weakly connected or almost isolated. Some papers are also connected to the main cluster through only a few links, showing that they are related but less central to the topic.

The map also helped identify papers that were not in the original reference list. Some notable ones include Brown (2020), Lewis (2020), Achiem (2023), Du (2023), Wang (2022),

Chen (2022), and Guo (2025). These papers provide additional connections to LLM development, reasoning, and research-idea generation. Overall, the map gives a clearer picture of which papers are central, peripheral, or isolated than the spreadsheet alone.